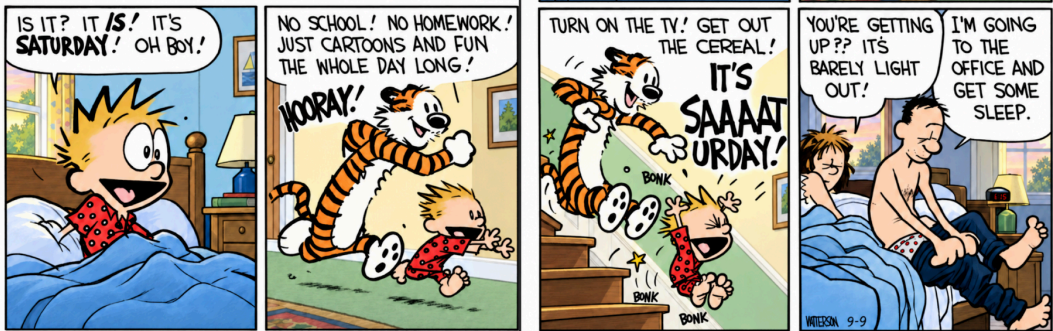
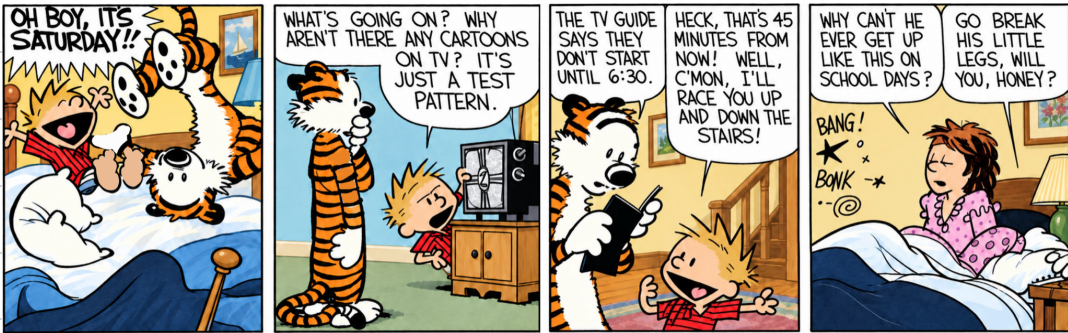




Memory Consumption by Language Models

Llama3.1:8B parameter model on my computer.

4.123423 -> 4.12

Inference:

$$\text{float16} = (8 * 10^9 * 2) / (1000 * 1000 * 1000) = 16\text{GB}$$

$$\text{int4} = (8 * 10^9 * 0.5) / (1000 * 1000 * 1000) = 4\text{GB}$$

Training: 8B

8B - floating16 -> 16 GB

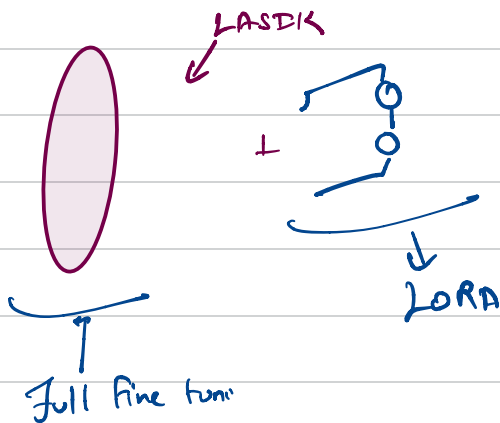
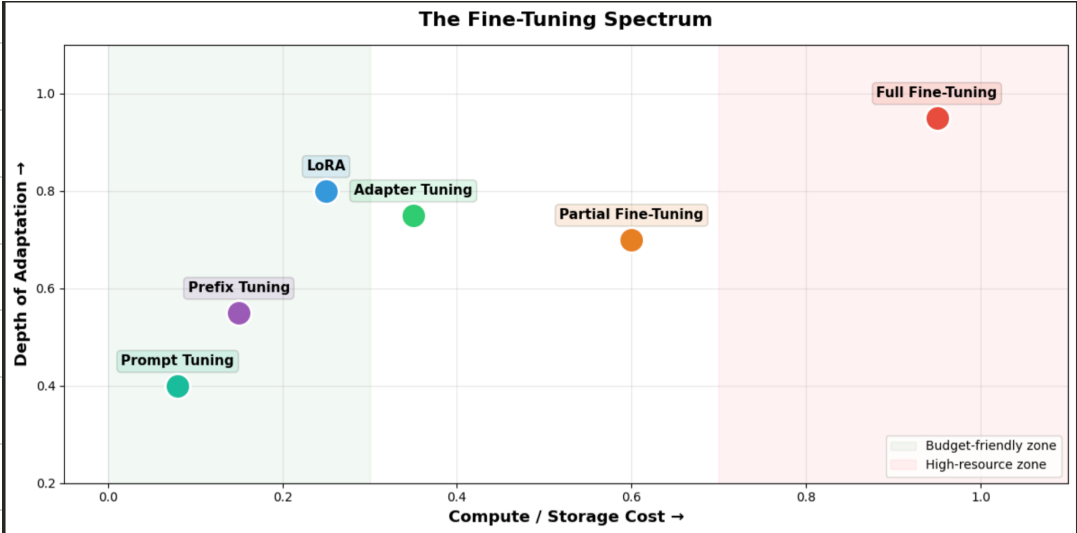
gradients 8B -> floating 16 -> 16 GB

Adam - M1 -> floating 32 -> 32 GB

M2 -> floating 32 -> 32 GB

$$16 + 16 + 32 + 32 = 112 \text{ GB VRAM}$$

Fine-Tuning Language Models



Inputs $\rightarrow []_{10 \times 100}$

Dense Layer 1 - 3000 neurons $\rightarrow []_{100 \times 3000}$



Output $[]_{10 \times 3000}$

Dense Layer 2 - 3000 $W = []_{3000 \times 3000}$

R_1

gradients of
917 Per

LoRA

$W + \Delta W$

$\downarrow$
 W'

LoRA = Low Rank

Adaptation Matrices

hyperparameter = 8

$W = []_{3000 \times 3000}$

$\Delta W \rightarrow$ Two Low Rank matrices

$A = []_{3000 \times 8}$

$B = []_{8 \times 3000}$

$$\Delta w = A @ B$$

Randomly Initialized Values

9M Param

48K

$$w' =$$

$$w + \Delta w$$

Frozen

Trained on our dataset

* Worked for 15m 13s · done 10:05 AM

► accept edits on (shift+tab to cycle) · ← for agents

188236 tokens
~188k uncached · /clear to start fresh
new task? /clear to save 189.2k tokens
/rc

Opus-5

Write

a

Pseudo

code

Inversion
sort

17m

Opus - 5

Write

a

Pseudo

code

Inversion
sort

Item

is

↑

Response-only masking for chat SFT

The model consumes the full conversation, but loss is computed only on the assistant target tokens.

